Arista Access Node JAY-ACC-55 Optical Transport Failure Mitigation SOP

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7750

Description

The Arista Access Node JAY-ACC-55 at the Jayanagar Metro Site is experiencing a critical optical transport failure, manifesting as a degradation of SFP/XFP modules, fiber cuts, and/or Rx/Tx power loss. This failure domain is specific to the Access role on Arista equipment due to the underlying protocol and hardware mechanics. In the context of Arista's Extensible Operating System (EOS), the Access role is responsible for facilitating data transmission and reception between the node and its peer nodes. The EOS, running on the Arista 7750 series switch, relies on the Optical Transport Network (OTN) protocol to manage the transmission and reception of optical signals. The OTN protocol ensures the reliable transmission of data over optical fibers by implementing features such as error correction, frame alignment, and framing. However, in the event of SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, the OTN protocol's ability to manage the optical signals is compromised, resulting in the critical optical transport failure observed on the Access role. This SOP aims to mitigate this failure domain by systematically diagnosing and resolving the root causes of the failure.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that remote hands are dispatched to the site.

- Check the status of redundant links to ensure that they are operational and can take over in case of a failure.

- Verify that the Arista 7750 series switch is powered on and that the system is stable.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware sfp, Expected Output: SFP information: slot 1: SFP-10G-SR, status: OK, Description: This command isolates the SFP/XFP module's status and verifies its operational integrity.

Command: show interface brief, Expected Output: Interface 1/1: administrative status up, link status up, mtu 1514, Description: This command checks the interface's link status and verifies its operational integrity.
```

```
Command: show optics, Expected Output: Optics information: slot 1: Rx
power: -2.5 dBm, Tx power: 2.5 dBm, Description: This command checks
the optical signal's receive and transmit power levels and verifies
their operational integrity.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: install hardware sfp slot 1 SFP-10G-SR, Description: This step involves replacing the SFP/XFP module with a new one to resolve the degradation issue.

Step 2: Step: Check and repair fiber cuts, Command: show interface 1/1, Description: This step involves checking the interface's link status and verifying its operational integrity. If the link status is down, it needs to be repaired by reconnecting the fiber cables.

Step 3: Step: Adjust Rx/Tx power levels, Command: set optics rx-power slot 1 -2.0 dBm, Description: This step involves adjusting the optical signal's receive and transmit power levels to resolve the power loss issue.

Post-Check Verification

```
Command: show hardware sfp, Expected Output: SFP information: slot 1:
SFP-10G-SR, status: OK, Description: This command verifies the SFP/XFP
module's status and ensures its operational integrity.

Command: show interface brief, Expected Output: Interface 1/1:
administrative status up, link status up, mtu 1514, Description: This
command checks the interface's link status and verifies its
operational integrity.

Command: show optics, Expected Output: Optics information: slot 1: Rx
power: -2.0 dBm, Tx power: 2.0 dBm, Description: This command checks
the optical signal's receive and transmit power levels and verifies
their operational integrity.
```

Roll-back Procedures

- Step: Restore previous SFP/XFP module, Command: remove hardware sfp slot 1, Description: This step involves restoring the previous SFP/XFP module to resolve the issue.

- Step: Restore previous fiber configuration, Command: show interface 1/1, Description: This step involves restoring the previous fiber configuration to resolve the issue.

- Step: Restore previous Rx/Tx power levels, Command: set optics rx-power slot 1 -2.5 dBm, Description: This step involves restoring the previous Rx/Tx power levels to resolve the issue.